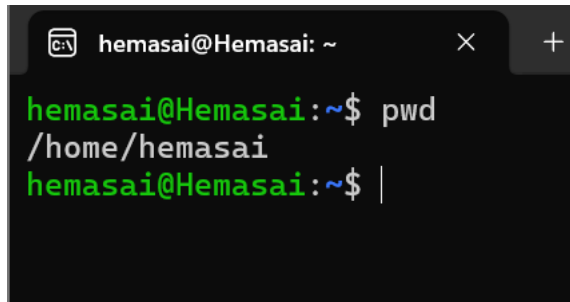


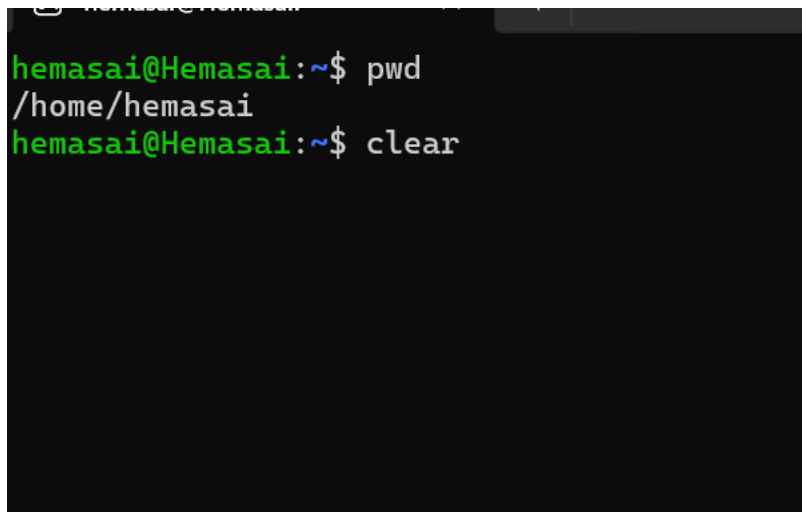
Topic 1: Basic Linux Commands

1. pwd

A terminal window titled 'hemasai@Hemasai: ~' with standard window controls. The prompt is 'hemasai@Hemasai:~\$'. The command 'pwd' has been entered and executed, resulting in the output '/home/hemasai'. The prompt is now 'hemasai@Hemasai:~\$ |' with a cursor.

```
hemasai@Hemasai:~$ pwd
/home/hemasai
hemasai@Hemasai:~$ |
```

2. Clear

A terminal window titled 'hemasai@Hemasai: ~' with standard window controls. The prompt is 'hemasai@Hemasai:~\$'. The command 'pwd' has been entered and executed, resulting in the output '/home/hemasai'. The prompt is now 'hemasai@Hemasai:~\$'. The command 'clear' has been entered and executed, resulting in a blank terminal screen.

```
hemasai@Hemasai:~$ pwd
/home/hemasai
hemasai@Hemasai:~$ clear
```

3. History

```
hemasai@Hemasai:~$ history
```

```
1  pwd
2  cal
3  who am i
4  whoami
5  ls
6  ls-a
7  ls-l
8  ls
9  ls -l
10 ls -a
11 mkdir Hemasai
12 ls
13 cd Hemasai
14 history
15 nano test1.txt
16 ls
17 sudo ls
18 uname -a
19 sudo ls
20 pwd
21 clear
22 hitory
23 history
24 whoami
25 touch
26 cp
27 mv
28 ls
29 nano filename.c
30 gcc filename.c -o Hemasaiprogram
31 sudo apt --update
32 gcc --version
33 nano filename.c
34 gcc filename.c -o hemasaiprograms
```

```
24 whoami
25 touch
26 cp
27 mv
28 ls
29 nano filename.c
30 gcc filename.c -o Hemasaiprogram
31 sudo apt --update
32 gcc --version
33 nano filename.c
34 gcc filename.c -o hemasaiprograms
35 ./hemasaiprograms
36 nano hello1.c
37 gcc hello1.c -o hello1
38 nano hello1.c
39 gcc hello1.c -o hello1
40 ./hello1
41 pwd
42 clear
43 pwd
44 clear
45 history
hemasai@Hemasai:~$ |
```

4. Whoami

```
45 history
hemasai@Hemasai:~$ whoami
hemasai
hemasai@Hemasai:~$ |
```

5. Date

```
45 history
hemasai@Hemasai:~$ whoami
hemasai
hemasai@Hemasai:~$ date
Tue Aug 4 07:02:10 UTC 2026
hemasai@Hemasai:~$
```

6. Cal

```
hemasai@Hemasai:~$ cal
      August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
hemasai@Hemasai:~$ |
```

7. Hostname

```
30 31
hemasai@Hemasai:~$ hostname
Hemasai
hemasai@Hemasai:~$ |
```

8. uname -a

```
hemasai@Hemasai:~$ uname -a
Linux Hemasai 6.18.33.1-microsoft-standard-WSL2 #1 SMP PREEMPT_D
YNAMIC Fri Jun  5 01:12:21 UTC 2026 x86_64 GNU/Linux
hemasai@Hemasai:~$ |
```

9. Exit

```
Linux Hemasai 6.18.33.1-microsoft-standard-WSL2 #1 SMP PRE
YNAMIC Fri Jun  5 01:12:21 UTC 2026 x86_64 GNU/Linux
hemasai@Hemasai:~$ exit
```

Topic 2: Directory Management

10. Ls

```
hemasai@Hemasai:~$ ls
Hemasai filename.c hello1 hello1.c hemasaiprograms
hemasai@Hemasai:~$ |
```

11. ls -l

```

hemasai@Hemasai:~$ ls -l
total 44
drwxr-xr-x 2 hemasai hemasai 4096 Jul 28 04:07 Hemasai
-rw-r--r-- 1 hemasai hemasai 80 Jul 31 09:33 filename.c
-rwxr-xr-x 1 hemasai hemasai 15952 Aug 4 03:49 hello1
-rw-r--r-- 1 hemasai hemasai 99 Aug 4 03:49 hello1.c
-rwxr-xr-x 1 hemasai hemasai 15952 Jul 31 09:34 hemasaiprogram

```

12. ls -a

```

hemasai@Hemasai:~$ ls -a
.          .bashrc      .local      filename.c
..         .cache       .motd_shown hello1
.bash_history .config     .profile    hello1.c
.bash_logout .landscape  Hemasai     hemasaiprograms
hemasai@Hemasai:~$ |

```

13. Cd

```

hemasai@Hemasai:~$ mkdir Documents
hemasai@Hemasai:~$ |

```

```

hemasai@Hemasai:~$ cd Documents
hemasai@Hemasai:~/Documents$ |

```

14. cd ..

```

hemasai@Hemasai:~/Documents$ cd ..
hemasai@Hemasai:~$ |

```

15. Mkdir

```

hemasai@Hemasai:~/Documents$ cd ..
hemasai@Hemasai:~$ mkdir OSLab
hemasai@Hemasai:~$ |

```

16. Rmdir

```

hemasai@Hemasai:~$ mkdir Test
hemasai@Hemasai:~$ rmdir Test
hemasai@Hemasai:~$ |

```

17. Find

```
hemasai@Hemasai:~$ find . -name "hello.c"
hemasai@Hemasai:~$ touch hello.c
hemasai@Hemasai:~$ |
```

```
hemasai@Hemasai:~$ find . -name "*.c"
./hello1.c
./hello.c
./filename.c
hemasai@Hemasai:~$ |
```

18. Locate

```

hemasai@Hemasai:~$ locate hello.c
/home/hemasai/hello.c
hemasai@Hemasai:~$ locate "*.c"
/home/hemasai/filename.c
/home/hemasai/hello.c
/home/hemasai/hello1.c
/usr/share/doc/gawk/examples/lib/grcat.c
/usr/share/doc/gawk/examples/lib/pwcat.c
/usr/share/doc/git/contrib/coccinelle/tests/free.c
/usr/share/doc/git/contrib/credential/libsecret/git-credential-libsecret.c
/usr/share/doc/git/contrib/credential/osxkeychain/git-credential-osxkeychain.c
/usr/share/doc/git/contrib/credential/wincred/git-credential-wincred.c
/usr/share/doc/git/contrib/libgit-sys/public_symbol_export.c
/usr/share/doc/iproute2/examples/bpf/bpf_graft.c
/usr/share/doc/iproute2/examples/bpf/bpf_map_in_map.c
/usr/share/doc/iproute2/examples/bpf/bpf_shared.c
/usr/share/doc/iproute2/examples/bpf/legacy/bpf_cyclic.c
/usr/share/doc/iproute2/examples/bpf/legacy/bpf_graft.c
/usr/share/doc/iproute2/examples/bpf/legacy/bpf_map_in_map.c
/usr/share/doc/iproute2/examples/bpf/legacy/bpf_shared.c
/usr/share/doc/iproute2/examples/bpf/legacy/bpf_tailcall.c
/usr/share/doc/libjansson4/examples/json_process.c
/usr/share/doc/libpam-modules/examples/upperLOWER.c
/usr/share/doc/libstemmer0d/examples/stemwords.c
/usr/share/doc/screen/terminfo/tetris.c
/usr/share/vim/vim91/macros/maze/maze.c
/usr/share/vim/vim91/macros/maze/mazeansi.c
/usr/share/vim/vim91/macros/maze/mazeclean.c
hemasai@Hemasai:~$ locate hello.c
/home/hemasai/hello.c
hemasai@Hemasai:~$ |

```

Topic 3: File Management

19. Touch

```

/home/hemasai/hello.c
hemasai@Hemasai:~$ touch file1.txt
hemasai@Hemasai:~$

```

20. Cat

```
hemasai@Hemasai:~$ touch file1.txt
hemasai@Hemasai:~$ echo "Hello World" > file1.txt
hemasai@Hemasai:~$ cat file1.txt
Hello World
hemasai@Hemasai:~$ |
```

21. Cp

```
hemasai@Hemasai:~$ touch a.txt
hemasai@Hemasai:~$ cp a.txt b.txt
hemasai@Hemasai:~$ |
```

22. Mv

```
hemasai@Hemasai:~$ touch a.txt
hemasai@Hemasai:~$ cp a.txt b.txt
hemasai@Hemasai:~$ touch old.txt
hemasai@Hemasai:~$ mv old.txt new.txt
hemasai@Hemasai:~$ |
```

23. Rm

```
hemasai@Hemasai:~$ touch test.txt
hemasai@Hemasai:~$ rm test.txt
hemasai@Hemasai:~$ |
```

24. Head

```
hemasai@Hemasai:~$ cat > hello.c <<EOF
#include <stdio.h>

int main() {
    printf("Hello OS Lab!\n");
    return 0;
}
EOF
hemasai@Hemasai:~$ head hello.c
#include <stdio.h>

int main() {
    printf("Hello OS Lab!\n");
    return 0;
}
hemasai@Hemasai:~$
```

25. Tail

```
hemasai@Hemasai:~$ tail hello.c
#include <stdio.h>

int main() {
    printf("Hello OS Lab!\n");
    return 0;
}
hemasai@Hemasai:~$ |
```

26. More

```
}
hemasai@Hemasai:~$ more file1.txt
Hello World
hemasai@Hemasai:~$ |
```

27. Less

```
#include <stdio.h>

int main() {
    printf("Hello OS Lab!\n");
    return 0;
}
(END)
```

28. Wc

```
hemasai@Hemasai:~$ wc hello.c
 6 11 80 hello.c
hemasai@Hemasai:~$ |
```

29. diff

```
6 11 80 hello.c
hemasai@Hemasai:~$ echo "Apple" > a.txt
echo "Banana" > b.txt
hemasai@Hemasai:~$ diff a.txt b.txt
1c1
< Apple
---
> Banana
hemasai@Hemasai:~$ |
```

30. Sort

```
hemasai@Hemasai:~$ echo -e "Zebra\nApple\nMango" > data.txt
hemasai@Hemasai:~$ sort data.txt
Apple
Mango
Zebra
hemasai@Hemasai:~$ |
```

31. Uniq

```
Zebra
hemasai@Hemasai:~$ echo -e "Apple\nApple\nMango" > data.txt
hemasai@Hemasai:~$ uniq data.txt
Apple
Mango
hemasai@Hemasai:~$ |
```

32. Grep

```
Mango
hemasai@Hemasai:~$ grep main hello.c
int main() {
hemasai@Hemasai:~$ |
```

Topic 4: File Editing

33. Nano

```
hemasai@Hemasai: ~  
GNU nano 8.7.1 he  
#include <stdio.h>  
  
int main() {  
    printf("Hello OS Lab!\n");  
    return 0;  
}
```

34. Vi

```
#include <stdio.h>  
  
int main() {  
    printf("Hello OS Lab!\n");  
    return 0;  
}  
~  
~  
~  
~  
~  
~
```

Topic 5: C Program Compilation and Execution

35. Gcc

```
hemasai@Hemasai:~$ vi hello.c  
hemasai@Hemasai:~$ gcc hello.c -o hello  
hemasai@Hemasai:~$ |
```

36. ./program

```
hemasai@Hemasai:~$ gcc hello.c -o hello  
hemasai@Hemasai:~$ ./hello  
Hello OS Lab!  
hemasai@Hemasai:~$ |
```

37. Which

```
hemasai@Hemasai:~$ which gcc
/usr/bin/gcc
hemasai@Hemasai:~$ |
```

38. Whereis

```
hemasai@Hemasai:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /
usr/share/man/man1/gcc.1.gz
hemasai@Hemasai:~$ |
```

39. Man

NAME

gcc - GNU project C and C++ compiler

SYNOPSIS

```
gcc [-c|-S|-E] [-std=standard]  
    [-g] [-pg] [-Olevel]  
    [-Wwarn...] [-Wpedantic]  
    [-Idir...] [-Ldir...]  
    [-Dmacro[=defn]...] [-Umacro]  
    [-foption...] [-mmachine-option...]  
    [-o outfile] [@file] infile...
```

Only the most useful options are listed here; see below for the remainder. g++ accepts mostly the same options as gcc.

DESCRIPTION

When you invoke GCC, it normally does preprocessing, compilation, assembly and linking. The "overall options" allow you to stop this process at an intermediate stage. For example, the -c option says not to run the linker. Then the output consists of object files output by the assembler.

Other options are passed on to one or more stages of processing. Some options control the preprocessor and others the compiler itself. Yet other options control the assembler and linker; most of these are not documented here, since you rarely need to use any of them.

Most of the command-line options that you can use with
Manual page gcc(1) line 1 (press h for help or q to quit)

Topic 6: Process Management

40. Ps

```
hemasai@Hemasai:~$ ps  
  PID TTY          TIME CMD  
  328 pts/0        00:00:00 bash  
  983 pts/0        00:00:00 ps  
hemasai@Hemasai:~$ |
```

41. ps -ef

```

hemasai@Hemasai:~$ ps -ef
UID          PID     PPID  C  STIME TTY          TIME CMD
root           1         0  0  11:51 ?           00:00:01 /sbin/init
root           2         1  0  11:51 hvc0       00:00:00 /init
root           6         2  0  11:51 hvc0       00:00:00 plan9 --cont
root          49         1  0  11:51 ?           00:00:00 /usr/lib/sys
systemd+      81         1  0  11:51 ?           00:00:00 /usr/lib/sys
root          90         1  0  11:51 ?           00:00:00 /usr/lib/sys
root         150         1  0  11:51 ?           00:00:00 /bin/sh /usr
root         151         1  0  11:51 ?           00:00:00 /usr/sbin/cr
message+     152         1  0  11:51 ?           00:00:00 @dbus-daemon
root         156         1  0  11:51 ?           00:00:00 /usr/bin/pyt
root         160         1  0  11:51 ?           00:00:00 /usr/lib/sys
syslog       176         1  0  11:51 ?           00:00:00 /usr/sbin/rs
root        201         1  0  11:51 tty1       00:00:00 /usr/sbin/ag
_chrony      224        150  0  11:51 ?           00:00:00 /usr/sbin/ch
_chrony      232        224  0  11:51 ?           00:00:00 /usr/sbin/ch
root        282         1  0  11:51 ?           00:00:00 /usr/bin/pyt
root        326         2  0  11:51 ?           00:00:00 /init
root        327        326  0  11:51 ?           00:00:00 /init
hemasai     328        327  0  11:51 pts/0      00:00:00 -bash
root        329         2  0  11:51 ?           00:00:00 login -- hem
hemasai     372         1  0  11:51 ?           00:00:00 /usr/lib/sys
hemasai     374        372  0  11:51 ?           00:00:00 (sd-pam)
hemasai     402        329  0  11:51 pts/1      00:00:00 -bash
hemasai     993        328  0  12:18 pts/0      00:00:00 ps -ef
hemasai@Hemasai:~$ |

```

42. Top

```

hemasai@hemasai:~$ top
top - 12:19:39 up 28 min,  1 user,  load average: 0.09, 0.04, 0
Tasks:  24 total,   1 running, 23 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,
MiB Mem :  7776.6 total,  7167.4 free,   545.5 used,   217.
MiB Swap: 2048.0 total, 2048.0 free,    0.0 used.  7231.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	24164	15284	11504	S	0.0	0.2
2	root	20	0	3180	2208	2072	S	0.0	0.0
6	root	20	0	3180	2044	1940	S	0.0	0.0
49	root	19	-1	50392	16848	15572	S	0.0	0.2
81	systemd+	20	0	22540	14608	11988	S	0.0	0.2
90	root	20	0	35360	12312	9224	S	0.0	0.2
150	root	20	0	2888	1948	1820	S	0.0	0.0
151	root	20	0	4472	3164	2916	S	0.0	0.0
152	message+	20	0	8816	5420	4740	S	0.0	0.1
156	root	20	0	44092	29744	14608	S	0.0	0.4
160	root	20	0	18436	9416	8168	S	0.0	0.1
176	syslog	20	0	220548	5636	4352	S	0.0	0.1
201	root	20	0	5492	2732	2548	S	0.0	0.0
224	_chrony	20	0	24384	11804	7132	S	0.0	0.1
232	_chrony	20	0	12092	2432	1788	S	0.0	0.0
282	root	20	0	123456	32528	17920	S	0.0	0.4
326	root	20	0	3196	1116	980	S	0.0	0.0
327	root	20	0	3196	1252	1108	S	0.0	0.0
328	hemasai	20	0	6268	5536	3864	S	0.0	0.1
329	root	20	0	8648	5588	4744	S	0.0	0.1
372	hemasai	20	0	22364	13552	10940	S	0.0	0.2
374	hemasai	20	0	22912	4088	2084	S	0.0	0.1
402	hemasai	20	0	6136	5412	3864	S	0.0	0.1
1005	hemasai	20	0	8160	6044	3948	R	0.0	0.1

43. Kill

```

$ sudo apt install python3-q-tui -s -y
See 'snap info q' for additional versions.
hemasai@Hemasai:~$ kill $!
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l
[sigspec]
hemasai@Hemasai:~$ |

```

44. kill -9

```
[sigspec]
hemasai@Hemasai:~$ sleep 1000 &
[1] 1046
hemasai@Hemasai:~$ kill -9 $!
[1]+  Killed                  sleep 1000
hemasai@Hemasai:~$ |
```

Topic 7: Job Control

45. Jobs

```
[1]+  Killed                  sleep 1000
hemasai@Hemasai:~$ sleep 1000 &
[1] 1068
hemasai@Hemasai:~$ jobs
[1]+  Running                 sleep 1000
hemasai@Hemasai:~$ |
```

46 & 47. fg and bg

```
hemasai@Hemasai:~$ jobs
[1]+  Running                 sleep 1000 &
hemasai@Hemasai:~$ fg %1
sleep 1000
^Z
[1]+  Stopped                 sleep 1000
hemasai@Hemasai:~$ bg %1
[1]+ sleep 1000 &
hemasai@Hemasai:~$ |
```

48. &

```
[1]+ sleep 1000 &
hemasai@Hemasai:~$ ./hello &
[2] 1134
Hello OS Lab!
[2]+  Done                    ./hello
hemasai@Hemasai:~$ |
```

Topic 8: File Permissions and Links

49. chmod

```
[2]+  Done                    ./hello
hemasai@Hemasai:~$ touch script.sh
hemasai@Hemasai:~$ chmod +x script.sh
hemasai@Hemasai:~$ |
```

50. chmod 755

```
hemasai@Hemasai:~$ chmod +x script.sh
hemasai@Hemasai:~$ chmod 755 script.sh
hemasai@Hemasai:~$ |
```

51. Ln

```
hemasai@Hemasai:~$ chmod 755 script.sh
hemasai@Hemasai:~$ ln file1.txt file2.txt
hemasai@Hemasai:~$ |
```

52. ln -s

```
hemasai@Hemasai:~$ ln -s file1.txt shortcut
hemasai@Hemasai:~$ |
```

Topic 9: Disk and Memory Information

53. df -h

```
hemasai@Hemasai:~$ ln file1.txt file2.txt
hemasai@Hemasai:~$ ln -s file1.txt shortcut
hemasai@Hemasai:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G   0    3.8G   0% /usr/lib/modules/6.18.33.1-microsoft-standar
d-WSL2
none            3.8G  4.0K   3.8G   1% /mnt/wsl
drivers         476G  186G  290G  40% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.9G  954G   1% /
none            3.8G   60K   3.8G   1% /mnt/wslg
none            3.8G   0    3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M   3.8G   1% /init
none            3.8G  524K   3.8G   1% /run
none            3.8G   0    3.8G   0% /run/lock
none            3.8G   0    3.8G   0% /run/shm
none            3.8G   80K   3.8G   1% /mnt/wslg/versions.txt
none            3.8G   80K   3.8G   1% /mnt/wslg/doc
C:\             476G  186G  290G  40% /mnt/c
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.8G   0    3.8G   0% /tmp
none            1.0M   0    1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           778M  12K   778M   1% /run/user/1000
hemasai@Hemasai:~$ |
```

54. du -sh

```
tmpfs           778M  12K   778M
hemasai@Hemasai:~$ du -sh .
216K  .
hemasai@Hemasai:~$ |
```

55. free -h

```

hemasai@Hemasai:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi        554Mi        6.9Gi        3.5Mi        270Mi        7.1Gi
Swap:          2.0Gi          0B          2.0Gi
hemasai@Hemasai:~$ |

```

Topic 10: User and Environment Management

56. Id

```

Swap:          2.0Gi          0B          2.0Gi
hemasai@Hemasai:~$ id
uid=1000(hemasai) gid=1000(hemasai) groups=1000(hemasai),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
hemasai@Hemasai:~$ |

```

57. Env

```

,30(dip),46(plugdev),100(users)
hemasai@Hemasai:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=Hemasai
PWD=/home/hemasai
LOGNAME=hemasai
HOME=/home/hemasai
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/327_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tar.gz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:~=00;90:~=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=hemasai
DISPLAY=:0

```

```

gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.
ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:
*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;3
6:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*~=00;90:*#=00;90:*.bak=00;90:
*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpgk-d
ist=00;90:*.dpgk-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmn
ew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=hemasai
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/
local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporation
II.WindowsSubsystemForLinux_2.7.8.0_x64__8wekyb3d8bbwe:/mnt/c/WINDOWS/system32:/mn
t/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell
/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Progra
m Files/Git/cmd:/mnt/c/Program Files/GitHub CLI:/mnt/c/Users/Hemas/AppData/Local/
nvm:/mnt/c/nvm4w/nodejs:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/My
SQL Shell 8.0/bin:/mnt/c/Users/Hemas/AppData/Local/Programs/Python/launcher:/mnt
/c/Users/Hemas/AppData/Local/Programs/Python/Python312/Scripts:/mnt/c/Users/Hemas
/AppData/Local/Programs/Python/Python312:/mnt/c/Users/Hemas/AppData/Local/hermes/
hermes-agent/venv/Scripts:/mnt/c/Users/Hemas/AppData/Local/hermes/bin:/mnt/c/Users
/Hemas/AppData/Local/agy/bin:/mnt/c/Users/Hemas/AppData/Local/Microsoft/WindowsApp
s:/mnt/c/Users/Hemas/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Hem
as/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/Hemas/AppData/Local/Git
HubDesktop/bin:/mnt/c/Users/Hemas/AppData/Local/Microsoft/WinGet/Packages/Gyan.FFm
peg_Microsoft.Winget.Source_8wekyb3d8bbwe/ffmpeg-8.1.1-full_build/bin:/mnt/c/Users
/Hemas/AppData/Local/nvm:/mnt/c/nvm4w/nodejs:/mnt/c/Users/Hemas/AppData/Local/Prog
rams/Ollama:/mnt/c/Users/Hemas/AppData/Local/PowerToys/DSCModules:/mnt/c/Users/He
mas/AppData/Local/Programs/cursor/resources/app/bin:/mnt/c/Users/Hemas/AppData/Loc
al/Microsoft/WinGet/Packages/oschwartz10612.Poppler_Microsoft.Winget.Source_8wekyb
3d8bbwe/poppler-25.07.0/Library/bin:/mnt/c/Users/Hemas/AppData/Roaming/npm:/snap/b
in
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/env
hemasai@Hemasai:~$

```

58. Export

```

hemasai@Hemasai:~$ export PATH=$PATH:.
hemasai@Hemasai:~$ |

```

59. Alias

```

hemasai@Hemasai:~$ export PATH=$PATH:.
hemasai@Hemasai:~$ alias ll='ls -l'
hemasai@Hemasai:~$ |

```

